

Adroit Technologies

Protocol Driver

Name	FAG SmartCheck WebService
DLL	D_FAG



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1. Overview

This driver allows communications to the FAG SmartCheck using its Webservice.

The driver supports only reading the alarms status and values of the following signals:

Velocity (iso10816_2hz_1khz_velocity)

Peak to Peak Acceleration (peak_2_peak_acceleration_high_vibration_values)

RMS Acceleration (rms_broadband_acceleration_overall_monitor)

RMS Demodulation (rms_broadband_demodulation_overall_monitor)

Temperature (temperature)



PLEASE NOTE

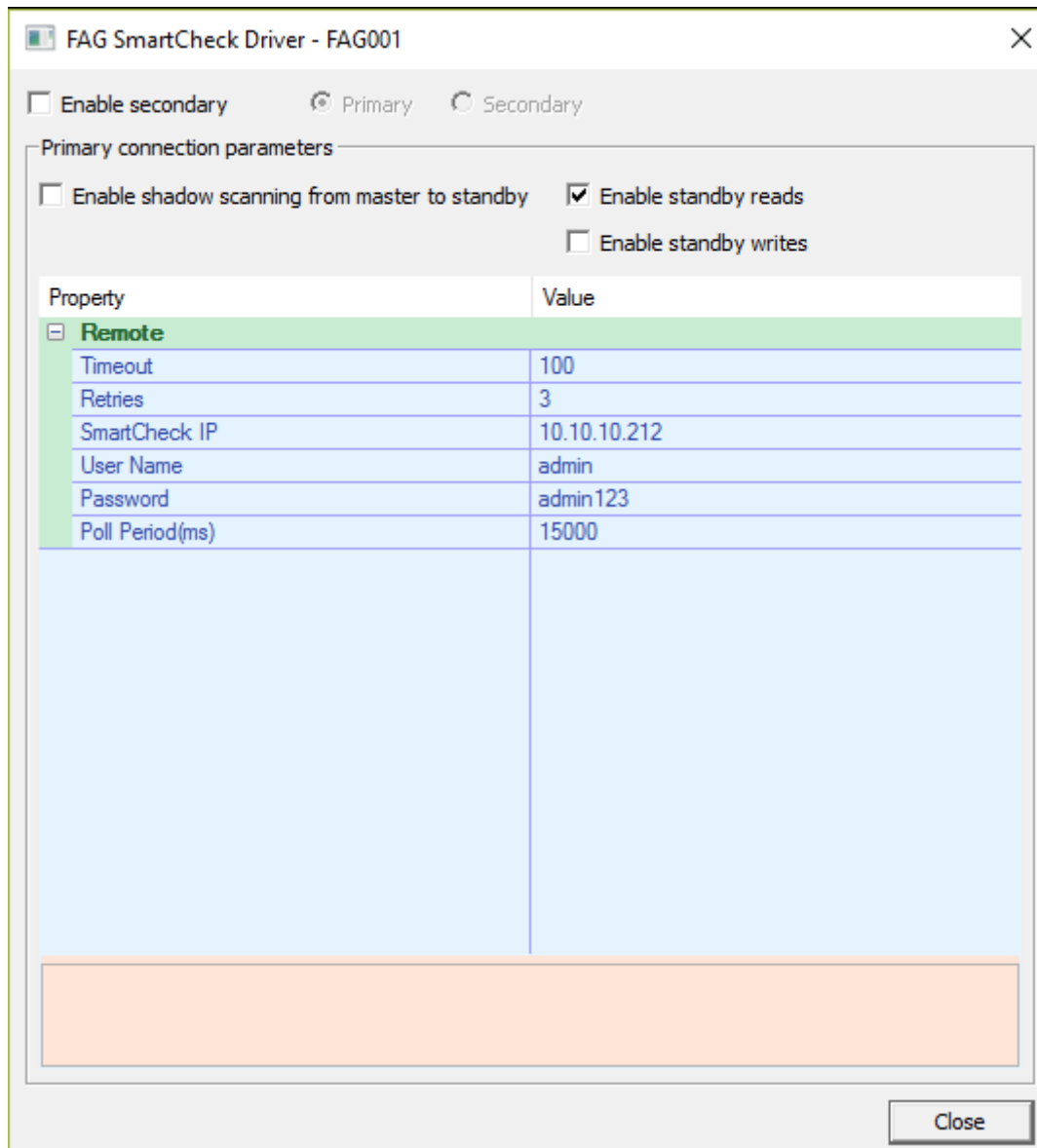
The FAG SmartCheck cannot accept communications quicker than 1 poll every 15 seconds and will be defaulted to 15 seconds. Nothing lower will be accepted and will be replaced with 15 seconds. For this reason, the actual time specified for each scan point is ignored.

2. Wiring

Ethernet

3. Device Configuration

3.1. Device Configuration



FAG SmartCheck Driver - FAG001

☐ Enable secondary ☒ Primary ☐ Secondary

Primary connection parameters

☐ Enable shadow scanning from master to standby ☒ Enable standby reads
☐ Enable standby writes

Property	Value
Remote	
Timeout	100
Retries	3
SmartCheck IP	10.10.10.212
User Name	admin
Password	admin123
Poll Period(ms)	15000

Close

3.1.1. Driver Details

Enable Secondary – Allows enabling of redundant device connections such that in case of a failure a switchover will occur. (Default = OFF)

Primary – Allows configuration setting of the Primary device.

Secondary – Allows configuration setting of the Secondary device.

3.1.2. Primary Connection Parameters

Enable shadow scanning from master to standby – If enabled will stop any communication to the device when the Agent server is in standby mode. By default we assume parallel scanning (default = OFF)

Enable standby reads – Allows reading from the device when agent server is in standby mode.
(Default = ON)

Enable standby writes – Allows writing to the device when agent server is in standby mode.
(Default = OFF)

Timeout – Time in milliseconds to wait for a response from the device. (Default = 1000 ms)

Retries – The maximum number of retries to attempt communication to the device before ensuring a complete failure. (Default = 3)

SmartCheck IP – The FAG SmartCheck device Station IP.

User Name – The FAG SmartCheck device username to access the WebService. (Default = admin)

Password – The FAG SmartCheck device password to access the WebService. (Default = admin123)

Poll Period (ms) – Time in milliseconds to Poll for a response from the device. (Default = 15000 ms - This is also the minimum possible for the FAG SmartCheck)

4. Supported Addresses

Scan Addresses takes the form of V:[Measurement Job Name].[Measurement Signal Name] (The Value) or S:[Measurement Job Name].[Measurement Signal Name] (The Alarm Status) exactly as it is shown in the FAG SmartCheck configuration.

For example:

Out of the BOX, default configuration measurement job for a FAG SmartCheck is as follows with its 5 Measurement Characteristics and the corresponding scan address for it as specified below:

The Unit Status:-> S:UNIT_STATUS

The Unit Firmware Version:-> V:UNIT_STATUS

The Peak-Peak Acceleration Status\Value:->

S\V:DEFAULT_CONFIGURATION_BASE_MEASUREMENT_JOB.default_configuration_peak_2_peak_acceleration_high_vibration_values

The RMS Broadband Acceleration Status\Value :->

S\V:DEFAULT_CONFIGURATION_BASE_MEASUREMENT_JOB.default_configuration_rms_broadband_acceleration_overall_monitor

The RMS Broadband Demodulation Status\Value :->

S\V:DEFAULT_CONFIGURATION_BASE_MEASUREMENT_JOB.default_configuration_rms_broadband_demodulation_overall_monitor

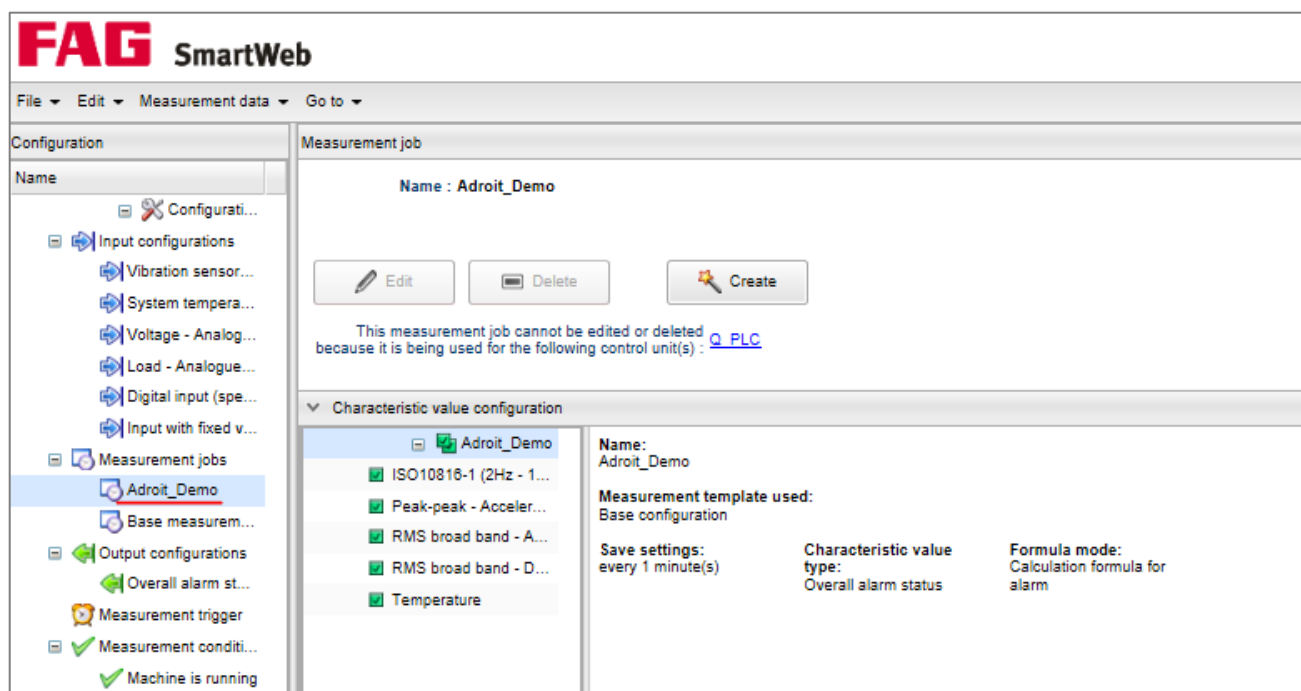
The Temperature Status\Value :->

S\V:DEFAULT_CONFIGURATION_BASE_MEASUREMENT_JOB.default_configuration_temperature

The Velocity Status\Value :->

S\V:DEFAULT_CONFIGURATION_BASE_MEASUREMENT_JOB.default_configuration_iso10816_2hz_1khz_velocity

If you want to address your own created Measurement Job as in the following image, where it is called "Adroit_Demo" the addressing will be:



The Unit Status:-> S:unit_status

The Temperature Status\Value :-> S\V:Adroit_Demo.Temperature

The Peak-Peak Acceleration Status\Value:-> S\V:Adroit_Demo.Peak-peak - Acceleration (High vibration values)

The RMS Broadband Acceleration Status\Value :-> V\S:Adroit_Demo.RMS broad band - Acceleration (Overall status)

The RMS Broadband Demodulation Status\Value :-> V\S:Adroit_Demo.RMS broad band - Demodulation (Overall status)

The Velocity Status\Value :-> V\S:Adroit_Demo.ISO10816-1 (2Hz - 1kHz) - Velocity

The STATUS addresses in the above table returns INTEGER values as shown below:

- 0** - No characteristic value has yet been measured.
- 1** - No alarm exists.
- 2** - Pre-alarm exists.
- 3** - Main alarm exists.
- 4** - Error Occurred.
- 5** - Valid Value was returned in the Value Scan (V.0 – V.5)

The VALUES will return a 32 bit floating point value.

5. Protocol Definition and Considerations

5.1. Message Protocol

```

ASCII DATASCOPE for FAG001 on 10.10.10.212
(PAUSED)
(RESUME) Added item S:unit_status
Added item V:Adroit_Demo.Temperature
Added item V:Adroit_Demo.Peak-peak - Acceleration (High vibration values)
Added item S:Adroit_Demo.RMS broad band - Acceleration (Overall status)
Added item V:Adroit_Demo.RMS broad band - Acceleration (Overall status)
Added item S:Adroit_Demo.RMS broad band - Demodulation (Overall status)
Added item S:Adroit_Demo.Peak-peak - Acceleration (High vibration values)
Added item S:Adroit_Demo.ISO10816-1 (2Hz - 1kHz) - Velocity
Added item S:Adroit_Demo.Temperature
Added item V:Adroit_Demo.ISO10816-1 (2Hz - 1kHz) - Velocity
Added item V:Adroit_Demo.RMS broad band - Demodulation (Overall status)

Device Started

Reading System Status!

Updated ADROIT_DEMO.RMS BROAD BAND - DEMODULATION (OVERALL STATUS) with VALUE: 0.00
Updated ADROIT_DEMO.ISO10816-1 (2HZ - 1KHZ) - VELOCITY with ALARM STATUS: 1
Updated UNIT_STATUS with ALARM STATUS: 1
Updated ADROIT_DEMO.PEAK-PEAK - ACCELERATION (HIGH VIBRATION VALUES) with ALARM STATUS: 1
Updated ADROIT_DEMO.RMS BROAD BAND - DEMODULATION (OVERALL STATUS) with ALARM STATUS: 1
Updated ADROIT_DEMO.TEMPERATURE with VALUE: 34.00
Updated ADROIT_DEMO.ISO10816-1 (2HZ - 1KHZ) - VELOCITY with VALUE: 0.07
Updated ADROIT_DEMO.TEMPERATURE with ALARM STATUS: 1
Updated ADROIT_DEMO.RMS BROAD BAND - ACCELERATION (OVERALL STATUS) with ALARM STATUS: 1
Updated ADROIT_DEMO.PEAK-PEAK - ACCELERATION (HIGH VIBRATION VALUES) with VALUE: 0.02
Updated ADROIT_DEMO.RMS BROAD BAND - ACCELERATION (OVERALL STATUS) with VALUE: 0.00
(PAUSED)

```

6. General Notes and Troubleshooting

6.1. Contact Details

Email: support@adroit.co.za and drivers@adroit.co.za

Phone: +27 11 658-8100

Web: www.adroit.co.za

6.2. Adroit Technologies Knowledge Base

For additional information, please refer to our Knowledgebase website:

[Latest Drivers topics - Features, discussions, tips, tricks, questions, problems and feedback \(adroittechnologiesautomation.com\)](http://adroittechnologiesautomation.com)

7. Revisions

7.1. Document Revision

Rev.	Date	Author	Comment	Approval
1.0	14-Dec-2022	J. Duncan	New Format	D. Jordaan
1.1		K. Robinson	Grammar and spelling edits, alignment and general editing.	D. Jordaan
2.0				
3.0				
4.0				
5.0				
6.0				

7.2. DLL Revision

Rev.	Date	Changes
1	01-Aug-2017	First Release of Driver.
12	13-July-2022	Improved driver timing.

